Controlling chaos in fast optical systems

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In many cases of practical importance, it is desirable to stabilize chaotic systems by applying only small perturbations to some accessible system parameter. An efficient scheme for achieving such control was proposed by Ott, Grebogi, and Yorke (OGY) [1] and variations of the scheme have been used successfully to control the dynamics of lasers [2]. The key idea is to take advantage of the unstable states embedded in the attractor. As the system approaches the unstable state, the size of the feedback signal required to keep it there diminishes and its smallness is limited only by the noise level in the system. We describe alternative implementations of the OGY scheme that efficiently stabilize the unstable periodic orbits (UPO's) and unstable steady-states (USS's) of fast dynamical systems using small perturbations. The schemes are well suited for controlling instabilities in high-speed optical system such as diode lasers.

An important issue that must be addressed when designing a controller for stabilizing the dynamics of fast systems is the unavoidable time-lag between the sensing of the state of the system and applying the feedback signal. The time-lag arises from two sources: calculating the feedback signal; and propagating the signal through the feedback loop. Control of a chaotic system is not possible if the time-lag is large because the state of the system is uncorrelated with the feedback signal due to the exponential divergence of nearby trajectories. Typically, the time-lag must be less than $\sim 1/\lambda$, where λ is the largest positive Lyapunov exponent characterizing the chaotic system. To address this issue, we have devised new control algorithms that require minimal computation and can be implemented easily using analog electronics or passive optical components [3].

For applications that require periodic laser oscillations, efficient stabilization of the UPO's can be achieved by feedback of a continuous error signal

$$\epsilon(t) = \vec{c}_f \cdot [\vec{\xi}(t) - (1-R)\sum_{k=1}^{\infty} R^{k-1}\vec{\xi}(t-k\tau)] \quad , \qquad (1)$$

that is proportional to the difference between the present value of an accessible state variable $\vec{\xi}(t)$ and an infinite series of values of the state variable delayed by integral multiples of the period of the orbit τ, where $0\leq R \leq 1$, and $\vec{c}_f$ is a gain vector. We emphasize that $\epsilon(t)$ vanishes when the system is on the UPO since $\vec{\xi}(t) = \vec{\xi}(t-k\tau)$. *Thus, there is no power dissipated in the feedback loop when control is successful.* The case $R = 0$ corresponds to the scheme investigated by Pyragas [4] and Lu and Harrison [5], and it was used successfully to control UPO's in a laser [6]. The more general scheme ($R \neq 0$) can stabilize highly unstable orbits and it is capable of extending the domain of effective control significantly [3].

We are currently investigating all-optical and electronic implementations of the control scheme. In the optical scheme, the feedback signal is generated by reflecting a fraction of the optical field generated by the laser from a Fabry-Perot interferometer (note the similarity between Eq. 1 and the well-know expression for the amplitude of light reflected from a Fabry-Pérot interferometer). Control is initiated by injecting the filtered field into the laser. In the second scheme, the state variable is a photocurrent proportional to the intensity of the field generated by the laser and the feedback signal is generated by filtering the photocurrent with a simple analog circuit. Control is initiated by adjusting the pump rate about its nominal value with the feedback signal. We will discuss the veracity of these schemes when applied to a simple laser model and we will report on our progress on stabilizing the dynamics of chaotic diode lasers.

For applications that require stable laser oscillations, optimal control of the USS's can be achieved by adjusting the feedback loop such that $\tau \rightarrow 0$, $R \rightarrow 1$, with the ratio $(1-R)/\tau$ finite. Under these conditions, the feedback signal is determined approximately by the differential equation

$$\frac{d\epsilon}{dt} = -\gamma_f \epsilon + \vec{c}_f \cdot \frac{d\vec{\xi}(t)}{dt} \quad , \tag{2}$$

where $\gamma_f = (1-R)/\tau$. Equation 2 is closely related to the transmission of an high-pass filter and the amplitude of light reflected from a high-finesse Fabry-Pérot interferometer with a short mirror spacing. Again, we emphasize that $\epsilon(t)$ vanishes when the system is on the USS since $d\vec{\xi}(t)/dt = 0$. *Thus, there is no power dissipated in the feedback loop when control is successful.* Lu and Harrison [5] found that USS's of a laser could be stabilized theoretically using a feedback signal given by Eq. 1 with $R = 0$ for some values of $\vec{c}_f$ and τ; however, the general case when $R \neq 0$ was not addressed. We will discuss our progress on stabilizing the USS's of a chaotic diode laser.

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